

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 2nd Term Examination, 2022

EE 2213

(Power Electronics)

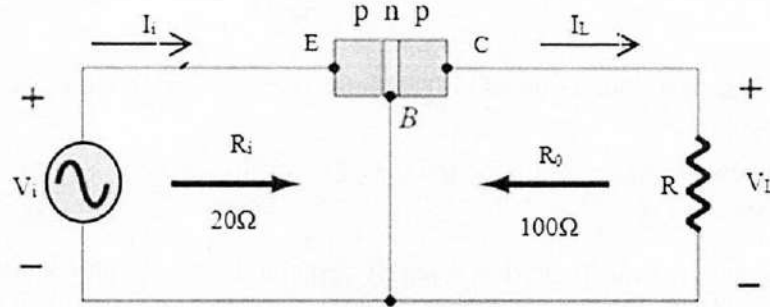
Time: 3 Hours.

Full Marks: 210

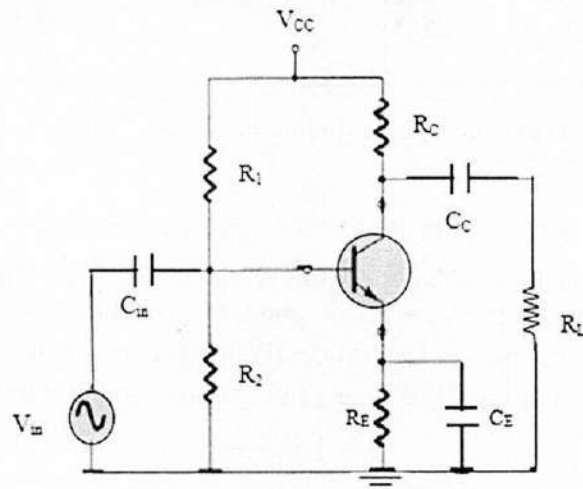
- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

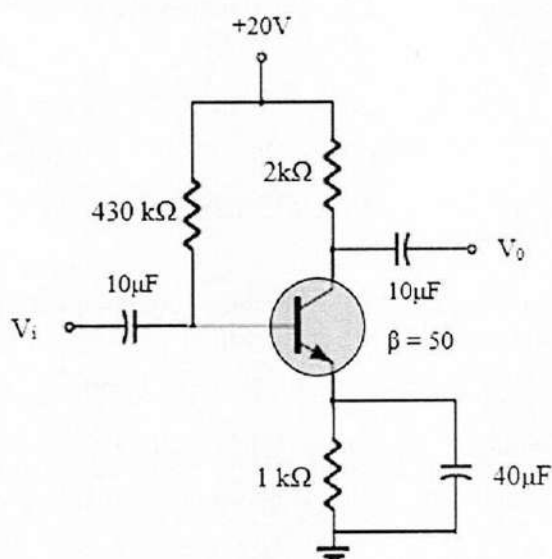
- 1(a). Draw the input and output characteristics graph of common emitter configuration and find out the reason of gradual shift of input graph with the increase of collector-emitter voltage. 12
- 1(b). Write down the necessary conditions for a transistor to work as an amplifier or switch. 05
- 1(c). For common emitter configuration, prove that $I_c = \beta I_B + (\beta + 1)I_{CBO}$. 10
- 1(d). Calculate the voltage gain for the network shown in figure if $V_i = 400$ mV and $R = 1$ k Ω . 08



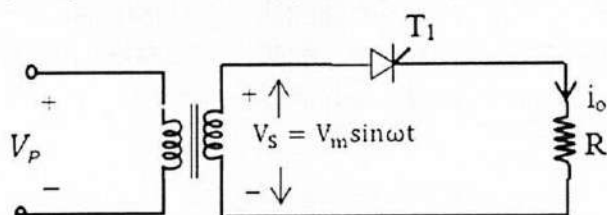
- 2(a). Derive the mathematical expressions of stability factor ($S(I_{CBO})$) for voltage divider bias configuration and hence show that voltage divider bias configuration is most stable. 15
- 2(b). Describe dc and ac load line for the following figure. 10



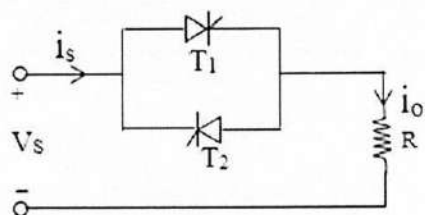
- 2(c). For the emitter bias network as shown in figure, determine: (i) I_C , (ii) V_C , (iii) V_{CE} , (iv) V_E , (v) V_{BC} . 10



- 3(a). Write down the basic operation of single-phase semi-converter with circuit diagram and necessary waveforms. 13
- 3(b). Draw the necessary wave-forms for the speed control of a dc motor by three-phase full converter system for $\alpha = \frac{\pi}{3}$. 12
- 3(c). If the converter shown in figure has a purely resistive load of R and delay angle is $\alpha = \frac{\pi}{2}$, determine (i) the rectification efficiency, (ii) the form factor, (iii) the ripple factor, (iv) the TUF and (v) the peak inverse voltage (PIV) of thyristor T_1 . 10

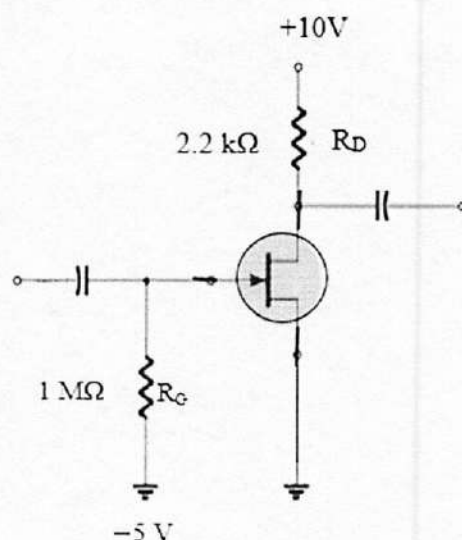


- 4(a). Explain the operation of three-phase bidirectional controller and draw the necessary waveforms for $\alpha = \frac{2\pi}{3}$. 17
- 4(b). Define power factor. Also describe the effects of low power factor. 08
- 4(c). A single-phase fullwave ac voltage controller shown in figure has a resistive load of $R = 10 \Omega$ and input voltage is $V_s = 120 \text{ V (rms)}$, 60 Hz. The delay angles of thyristors T_1 and T_2 are equal: $\alpha_1 = \alpha_2 = \alpha = \frac{\pi}{2}$. Determine (i) the rms output voltage V_o , (ii) the input PF, (iii) the average current of thyristors I_A and (iv) the rms current of thyristors I_R . 10



SECTION – B

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|-------|--|----|
| 5(a). | Distinguish between field effect transistor and bipolar junction transistor. | 07 |
| 5(b). | Briefly describe the constructional details of JFET with necessary diagrams. | 10 |
| 5(c). | What is meant by JFET biasing? What are the types of JFET biasing? | 10 |
| 5(d). | A JFET in following figure has values of $V_{GS(off)} = -8\text{ V}$ and $I_{DSS} = 16\text{ mA}$. Determine the values of V_{GS} , I_D and V_{DS} for the circuit. | 08 |



- | | | |
|-------|--|----|
| 6(a). | Why germanium is not used for controlled rectifier? | 05 |
| 6(b). | Briefly explain the thyristor turn-on methods. | 10 |
| 6(c). | What is di/dt protection and dv/dt protection in thyristor? How di/dt and dv/dt protection is achieved in thyristor? | 12 |
| 6(d). | An SCR full-wave rectifier supplies to a load of $100\ \Omega$. If the peak ac voltage between center tap and one end of secondary is 200 V , find (i) dc output voltage and (ii) load current for a firing angle of 60° . | 08 |
| 7(a). | Depending on the direction of current and voltage flows, write down the classification of DC converters and briefly discuss about four-quadrant converter. | 10 |
| 7(b). | Write down the differences between DIAC and TRIAC. | 08 |
| 7(c). | Why transistor is called latching device? | 05 |
| 7(d). | Power (brightness) of a 100 W , 110 V lamp is to be varied by controlling firing angle of SCR full-wave circuit, the rms value of ac voltage appearing across each SCR being 110 V . Find the rms voltage and current in the lamp at firing angle of 30° . | 12 |
| 8(a). | What is meant by inverter? Write down the applications of inverter. | 08 |
| 8(b). | Draw and explain the circuit diagram of three-phase inverter formed by single-phase inverters. | 12 |
| 8(c). | For a three-phase inverter, discuss and draw the circuit diagram with waveform for 180° conduction. | 15 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 2nd Term Examination, 2022

ESE 2209

(Bio and Wind Energy Engineering)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

ii) Figures in the right margin indicate full marks.

iii) Assume reasonable data if any missing.

SECTION – A

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|-------|--|----|
| 1(a). | How can water be removed from bioethanol? Explain. | 07 |
| 1(b). | Explain the formation of biomass energy through photosynthesis process with neat sketches. | 12 |
| 1(c). | What are the challenges and opportunities associated with the large-scale production of biomass energy? | 08 |
| 1(d). | How does the density of biomass briquettes impact their energy content and combustion efficiency? | 08 |
| 2(a). | What is biogas? What are the primary feed-stok's used in biogas production? | 06 |
| 2(b). | Discuss in detail the anaerobic digestion process, explaining the parameters that need to be controlled to achieve efficient biogas production. | 12 |
| 2(c). | Suppose a house owner has 3 cows and 4 buffalos. Each cow and buffalos is producing dung about 7 kg/day and 12 kg/day, respectively.
(i) By using the dung obtained from the above information in a biogas plant, how much energy can be produced per day?
(ii) Suppose the house owner has three bulbs of 40 watt and one fan of 80 watt. Is it possible to supply the required electrical energy needed for running the bulbs and fan from the biogas plant if 65% of the energy is lost during the transmission? [Gas production per kg of dung = 0.04 m ³] | 08 |
| 2(d). | Why is an acidic condition unfavorable for biogas production? Under what condition does ammonia accumulate in the biogas digester and what are the consequences of such accumulation? | 09 |
| 3(a). | What is pyrolysis of biomass? What are the main differences between fast pyrolysis and slow pyrolysis processes? | 07 |
| 3(b). | Write down the name of three pyrolysis processes. With a schematic diagram, explain bubbling fluidized bed purolysis reactor. | 10 |
| 3(c). | What is bioethanol? Briefly discuss the different growth phases of microorganism for bioethanol fermentation. | 10 |
| 3(d). | Explain the characteristics of a fuel. Compare the bio-fuel and fossil fuel. | 08 |
| 4(a). | What is COD? How COD of waste water sample can be measured? | 08 |
| 4(b). | Suppose a 12 mL waste water sample is taken for 7 days Biochemical Oxygen Demand (BOD) test. The sample is diluted with dilution water to a total volume of 275 mL. The initial Dissolved Oxygen (DO) concentration is 8 mg/L, and after 7 days, it has been reduced to 3 mg/L. Determine the BOD of the water sample and discuss the implications for the water quality. | 10 |
| 4(c). | What is biodiesel? Explain in details the steps involved in the production of biodiesel from vegetable oil. | 10 |
| 4(d). | Discuss the rate of catalyst in the transesterification process of biodiesel production. | 07 |

SECTION – B

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|--|----|
| 5(a). How does the relationship between air density and air velocity impact the performance of wind turbines? Discuss. | 07 |
| 5(b). Explain the working principle of a hot wire anemometer. Discuss specific applications where different kinds of anemometers are most commonly used. | 12 |
| 5(c). Why is wind data analysis crucial for the planning and optimization of wind energy project? In the context of wind energy, discuss the significance of the shape and scale parameters (k and c) in the Weibull distribution. | 11 |
| 5(d). The wind velocity measured at 15 m height at a meteorological observatory is 8 m/s. Find out the velocity at 50 m height at a wind turbine site having similar wind profile. The roughness heights at the observatory and wind turbine location are 0.03 m and 0.1 m, respectively | 05 |
| 6(a). Explain the significance of the power coefficient (C_P) and the torque coefficient (C_T) in the context of wind turbine aerodynamics. Prove that the maximum theoretical power coefficient of a horizontal axis wind turbine is $16/27$ and the maximum power produced is $P_{max} = \frac{1}{2} \rho_a A_T V^3 \frac{16}{27}$, where the symbols have their usual meaning. | 13 |
| 6(b). A 40 m diameter, three blade wind turbine produces 700 kW at a wind speed (hub height) of 14 m/s. The air density is 1.225 kg/m ³ . Find the followings:
(i) The rotational speed (rpm) of the rotor at a tip speed ratio of 5.00.
(ii) the tip-speed (m/s).
(iii) the efficiency of wind turbine system (including blades, transmission, shafts, and generator) under these conditions. | 07 |
| 6(c). How does blade design affect efficiency of wind turbine? | 08 |
| 6(d). Why is the gearbox an essential component in most wind turbines and how does its function contribute to the overall efficiency and performance of the turbine in converting wind energy into electricity | 07 |
| 7(a). Why is power system stability considered a crucial factor in ensuring the performance and reliability of wind turbines, and how does it impact their ability to contribute effectively to the overall energy grid? | 08 |
| 7(b). What are the key advantages of using a Doubly-Fed Induction Generator (DFIG) in a wind turbine, and how does its unique configuration contribute to improve efficiency and grid integration compared to other generator types? Discuss with schematic diagram. | 12 |
| 7(c). What are the purposes of reactive power control in wind turbines, and how does the ability to produce or consume reactive power improve the stability and effectiveness of the electrical grid? Explain. | 10 |
| 7(d). With the help of a diagram, explain how wind energy is converted into electric energy. | 05 |
| 8(a). What is the importance of project planning for wind power plant? Discuss the several steps for creating a project plan for a wind turbine power plant project. | 10 |
| 8(b). Explain the different ways in which the cost of a wind energy system is commonly expressed. | 07 |
| 8(c). How does the life cycle emission of wind turbines compare to those of traditional fossil fuel-based power generation? Discuss. | 08 |
| 8(d). How does a wind generator's project life affect its financial performance? Which procedure is used to integrate wind energy into the grid? | 10 |

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 2nd Term Examination, 2022

Hum 2213

(Economics and Accounting)

Time: 3 Hours.

Full Marks: 210

- N.B. i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION – A

- 1 Consider the following label and answer the questions below:

35

Price (\$/Bushel)	Demand for wheat (millions Bushel)	Supply of wheat (millions Bushel)
5	35	15
6	25	17
7	20	20
8	10	25
9	02	27

- (i) Define equilibrium price. Find out the equilibrium price and quantity?
(ii) Define price ceiling. If the government fixed a price ceiling of wheat per bushel at \$8, will there be surplus or shortage in the market? Show and explain it.
(iii) Define price floor. If the price floor is \$5, explain the market condition with the aid of diagram.
(iv) Explain the non-price determinants of supply.
- 2(a). Define (i) price elasticity of demand, and (ii) cross-price elasticity of demand. 15
(iii) List the determinants of price elasticity of demand with example.
- 2(b). Explain the problem of economic growth. On what factors does economic growth depend? Discuss briefly. 10
- 2(c). Distinguish between cardinal utility and ordinal utility. Which one is more realistic? Why? 10
- 3(a). Make a difference between GNP and GDP. 05
- 3(b). Define National income. Discuss the circular flow of national income. 15
- 3(c). What is demand-pull and cost-push inflation? Explain the main causes of inflation. 15
- 4(a). How to calculate GDP at market price, GNP, and national income? Discuss. 10
- 4(b). Define market. Distinguish between perfect competition and monopolistic competition market. 10
- 4(c). Show with the aid of graph, the short-run equilibrium of a firm in perfect competition. 15

SECTION – B

- 5(a). What is accounting? Who are the users of accounting information? Discuss. 10
- 5(b). Identify and describe the steps of accounting cycle. 10
- 5(c). Describe the basic rules of debit and credit of accounts. 15
- 6(a). Define cost and expense. Classify costs based on cost behavior. Explain with examples. 15
- 6(b). What are the limitations of trial balance? Discuss with example. 10
- 6(c). Suppose the acquire cost of an asset is tk. 55,000; salvage value at the end of 6 years is tk. 7,000. Calculate depreciation of the asset following straight-line method; and prepare a depreciation schedule following double-decline balance method. 10

7. Mr. Joy started Joy Travel Agency on 01 May 2023. The following activities occurred during the month:

May 01: Started services with cash tk. 42,000 and equipment tk. 65,000 funneled by Mr. Joy.
 May 05: Received from customers tk. 17,000 for providing services.
 May 07: Paid for internet and telephone bill tk. 5,000
 May 10: Purchased supplies on account tk. 3,000
 May 12: Took a bank loan tk. 75,000
 May 15: Paid for salaries tk. 10,000, while tk. 3,000 remained unpaid.
 May 19: Paid for 6-month insurance tk. 12,000
 May 22: Bad debt calculated tk. 4,200
 May 24: Paid 2/3 of the due amount against purchased credit supplies.
 May 26: Service provided but still unbilled tk. 35,000
 May 29: Received a bill from ABC Ad. Agency and hold payment for later date tk. 7,000
 May 30: Charged depreciation on equipment tk. 2,300

Required:

(a) Journalize the above transactions.

24

(b) Post the transactions (from May 15 to May 24) to ledger accounts.

11

8.

Salim Enterprise

35

Trial Balance as on December 31, 2022

Accounts title	Debit (tk.)	Accounts title	Credit (tk.)
Cash	5,000	Sales revenue	98,000
Equipment	50,000	Purchase return	2,000
Supplies	10,000	Purchase discount	1,000
Accounts receivable	24,000	Accounts payable	5,000
Inventory (01-01-2022)	8,000	Notes payable	10,000
Purchase	47,000	Capital	87,000
Freight-in	3,000	Bad debt allowances	5,000
Freight-out	2,000	Loan payable	5,000
Salaries expense	15,000		
Travel expense	3,000		
Rent expense	10,000		
Insurance (prepaid)	6,000		
Bad debt expense	3,000		
10% investment	20,000		
Drawings	5,000		
Rates and taxes	2,000		
Total	213,000	Total	213,000

Other data:

(i) Ending inventory was valued tk. 12,000.

(ii) Supplies on hand tk. 4,500.

(iii) Depreciation to be charged on equipment @10%.

(iv) Salaries outstanding tk. 3,500.

Required: (i) A comprehensive income statement

(ii) An owner's equity statement

(iii) A statement of financial positions.

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 2nd Term Examination, 2022

Math 2213

(Complex Variables and Fourier Analysis)

Time: 3 Hours.

Full Marks: 210

N.B. i) Answer any THREE questions from each section in separate scripts.

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SECTION – A

- 1(a). Represent $-5 + 5i$ and $-6 - \sqrt{2}i$ in polar form. Find the roots of $(-1 + i)^{1/3}$ and locate them graphically. 10
- 1(b). Prove an analytic function with constant modulus is constant. 07
- 1(c). Find the orthogonal trajectories of the family of curves in the xy -plane defined by $e^{-x}(x \sin y - y \cos y) = \alpha$, where α is a real constant. 12
- 1(d). Does the $\lim_{z \rightarrow 0} \frac{\bar{z}}{z}$ exist? 06
- 2(a). Define harmonic function. Prove that $u = 3x^2y + 2x^2 - y^3 - 2y^2$ is harmonic. Also find v such that $f(z) = u + iv$ is harmonic, where v is harmonic conjugate. 14
- 2(b). Evaluate $\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz$, where C is the circle $|z| = 3$. 11
- 2(c). Define isolated, essential, pole and removable singularities with example. 10
- 3(a). If $f(z) = u + iv$ is any analytic function of the complex variable z and $u - v = e^x(\cos y - \sin y)$, find $f(z)$ in terms of z . 14
- 3(b). Define Bessel's differential equation and its solution. 11
Prove that $J_{-n}(x) = (-1)^n J_n(x)$.
- 3(c). Prove that $J_{5/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{3}{x} \sin x + \frac{3-x^2}{x^2} \cos x \right)$. 10
- 4(a). Prove that $J_n(x) = \frac{1}{\pi} \int_0^\pi \cos(n\theta - x \sin \theta) d\theta$. 13
- 4(b). Prove that $\frac{d}{dx} \{x^n J_n(x)\} = x^n J_{n-1}(x)$. 08
- 4(c). If $f(z)$ is analytic at z_0 , prove that it must be continuous at z_0 . Give an example to show that the converse of $f(z)$ is not necessary true. 14

SECTION – B

- 5(a). Define periodic function. Find the period of the function $\sin 4\pi x$ and $\cos\left(\frac{n\pi x}{l}\right)$. Find the trigonometric Fourier series of the following function: 18
- $$f(x) = \begin{cases} 0, & -\pi < x < 0 \\ 2, & 0 < x < \pi \end{cases}$$
- 5(b). Develop $f(x)$ in Fourier series in the interval $(-2, 2)$ if $f(x) = 0$ for $-2 < x < 0$ and $f(x) = 1$ for $0 < x < 2$. 17
- 6(a). Explain the existence of Fourier series of a function. State and explain the physical meaning of Parseval's theorem. 18
- 6(b). Form a partial differential equation from $x^2 + y^2 + (z - c)^2 = a^2$. 11
- 6(c). Define linear and nonlinear partial differential equation. 06
- 7(a). Consider the temperature of a thin metal bar of length L of constant cross-section and homogeneous metal bar which is oriented along x -axis and perfectly insulated laterally. The ends of the bar are kept at temperature zero where the initial temperature is $f(x)$. Govern the heat equation and derive the temperature distribution formula. 20
- 7(b). Find the temperature $u(x, t)$ in a laterally insulated copper bar 70 cm long if the initial temperature is $100 \sin(3\pi x/70)^\circ\text{C}$ and the ends are kept at 0°C . How long will it take for the maximum temperature in the bar drop to 50°C . [density 8.92 g/cm^3 , specific heat $0.092 \text{ cal/(g}^\circ\text{C)}$, thermal conductivity $0.092 \text{ cal/(cm. sec. }^\circ\text{C)}$]. 15
- 8(a). Find the solution of the Laplace's equation in polar form. 12
- 8(b). Consider a circular membrane of radius R which is radially symmetric. The initial displacement and velocity of the membrane $f(r)$ and $g(r)$, respectively. Determine its solution $u(r, t)$. 23

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KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY

Department of Energy Science and Engineering

B. Sc. Engineering 2nd Year 2nd Term Examination, 2022

ME 2213

(Dynamics and Kinematics of Machineries)

Time: 3 Hours.

Full Marks: 210

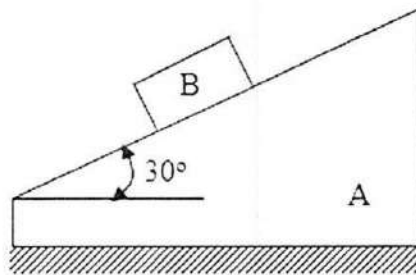
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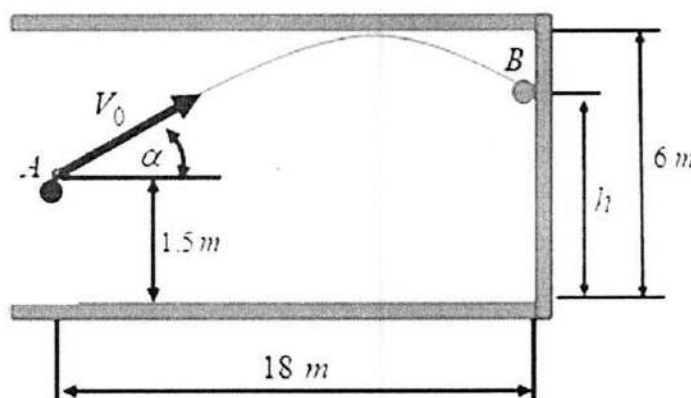
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SECTION – A

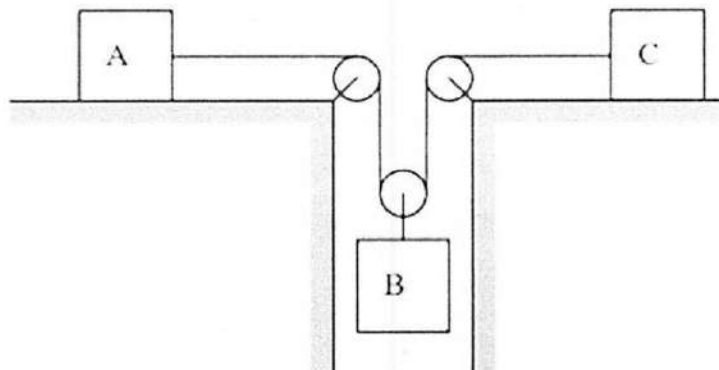
- 1(a). How does kinetics differ from kinematics? Why is study of dynamics needed as an energy science engineer? 07
- 1(b). At $t = 0$, wedge A starts moving to the left with a constant acceleration of 80 mm/s^2 and block B starts moving along the wedge towards the right with a constant acceleration of 120 mm/s^2 relative to the wedge. Determine (i) the acceleration of block B, (ii) the velocity of block B when $t = 3$ seconds. 13



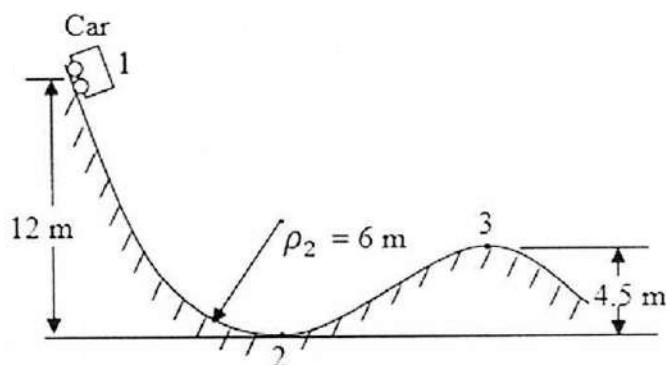
- 1(c). A player throws a ball with an initial velocity V_0 of 16 m/s from point A located 1.5 m above the floor. Knowing that $h = 3.5 \text{ m}$, determine the angle α for which the ball will strike the wall at point B. 16



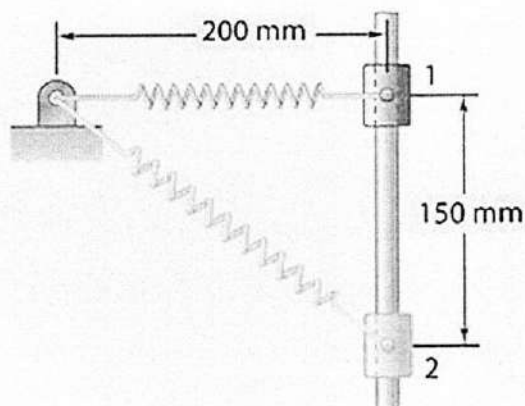
- 2(a). The coefficients of friction between block A and C and the horizontal surfaces are $\mu_s = 0.24$ and $\mu_k = 0.20$. If $m_A = 5 \text{ kg}$, $m_B = 10 \text{ kg}$, $m_C = 10 \text{ kg}$, determine: (i) the tension in the cord, (ii) the acceleration of each block. 18



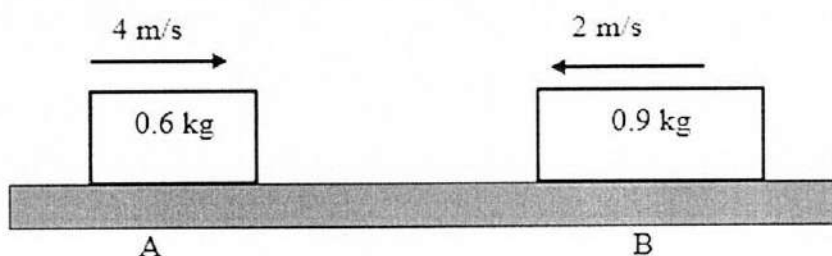
- 2(b). A 907 kg car starts from rest at point 1 and moves without friction down the track shown. 17
 (i) Determine the force exerted by the track on the car at point 2, where the radius of the curvature of the track is 6 m. (ii) Determine the minimum safe value of the radius of curvature at point 3.



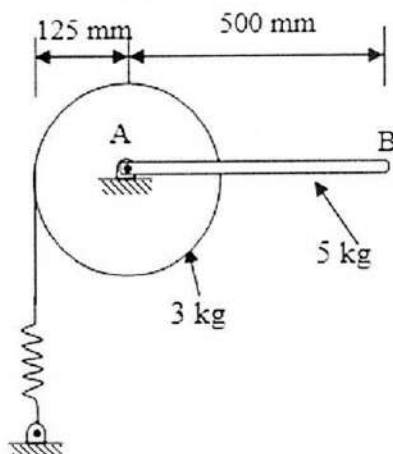
- 3(a). A 9 kg collar slides without friction along a vertical rod as shown. The spring attached to the collar has an undeformed length of 0.1 m and a constant of 0.9 kg/m. If the collar is released from rest in position 1, determine its velocity after it has moved 150 mm to position 2. 18



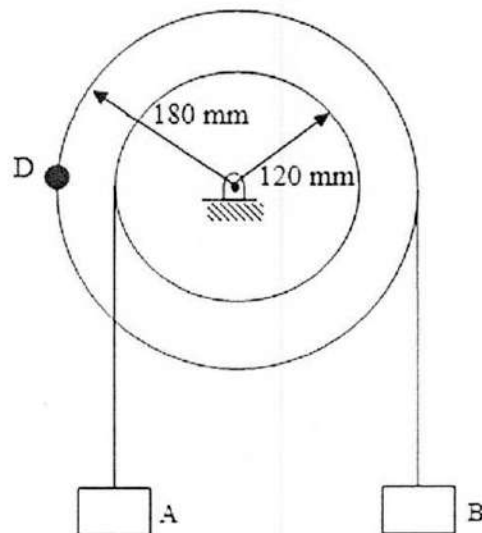
- 3(b). Two steel blocks are sliding on a frictionless horizontal surface with the velocity as shown. 17
 Knowing that coefficient of restitution $e = 0.75$, determine (i) the velocity of each block after impact, (ii) the loss of kinetic energy due to the impact.



- 4(a). The 5 kg slender rod AB is welded to the 3 kg uniform disk which rotates about a pivot at A. A 18
 spring of constant of 80 N/m is attached to the disk and is unstretched when the rod AB is horizontal. Knowing that the assembly is released from rest in the position shown, determine its angular velocity after it has rotated through 90° .

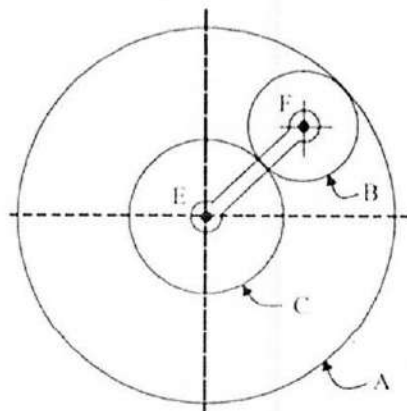


- 4(b). A pulley and two loads are connected by inextensible cords as shown. Load A has a constant acceleration of 300 mm/sec^2 and an initial velocity of 240 mm/sec , both directed upward. Determine (i) the number of revolutions executed by the pulley in 3 sec, (ii) the velocity and position of load B after 3 sec, (iii) the acceleration of point D on the rim of the pulley at $t = 0$. 17



SECTION – B

- 5(a). What is meant by static and dynamic balancing? Why reciprocating masses are only partially balanced? What are the effects of partial balancing of reciprocating parts in two-cylinder locomotives? 15
- 5(b). “For a multi-cylinder engine, the primary forces may be completely balanced by suitably arranging the crank angles, provided that the number of cranks are not less than four” – Why? 06
- 5(c). A two-cylinder uncoupled locomotive with cranks at 90° has a crank radius of 325 mm. The distance between the centers of driving wheels is 1.5 m. The pitch of the cylinders is 0.6 m. The diameter of treads of driving wheels is 1.8 m. The radius of center of gravity of balance masses is 0.65 m. The pressure due to dead load on each wheel is 40 kN. The masses of reciprocating and rotating parts per cylinder are 330 kg and 300 kg, respectively. The speed of the locomotive is 60 km/h. Find the followings: 14
- (i) The balancing masses both in magnitude and position required to be placed in the planes of driving wheels to balance whole of the revolving and two-third of the reciprocating masses
- (ii) The maximum speed at which it is possible to run the locomotive, in order that the wheels are not lifted from the rails.
- 6(a). State and explain the law of gearing with necessary equations and figures. 12
- 6(b). Draw schematic diagrams of four types of gear train. 05
- 6(c). An epicyclic gear train consists of three gears A, B and C as shown in figure. The gear A has 72 internal teeth and gear C has 32 internal teeth. Gear B meshes with both A and C and is carried on an arm EE which rotates about the center of A at 18 rpm. If the gear A is fixed, determine the speed of gears B and C. 18



- 7(a). Draw a 2D sketch of the following governors mentioning unique features of each one. 18
 (i) Watt, (ii) Porter, (iii) Proell, (iv) Hartnell, (v) Hartung, (vi) Pickering
- 7(b). A porter governor has arms 250 mm each and four rotating flyballs of mass 0.8 kg each. The sleeve movement is restricted to 20 mm either way from the height when the mean speed is 100 rpm. Calculate the central dead load and sensitivity of the governor neglecting friction when the flyball exerts a centrifugal force of 9.81 N. Also determine the effort and power for 3 percent speed change. 17
- 8(a). What is the significance of turning moment diagram of an internal combustion engine? 07
- 8(b). Differentiate between crank effort, crank-pin effort and piston effort. Show the forces acting on the reciprocating parts of an engine with appropriate diagram. [Neglect weight of the connecting rod] 10
- 8(c). A single cylinder, single acting, four stroke gas engine develops 20 kW at 300 rpm. The work done by the gases during the expansion stroke is three times the work done on the gases during the compression stroke, the work done during the suction and exhaust strokes being negligible. If the total fluctuation of speed is not to exceed $\pm 2\%$ of the mean speed and the turning moment diagram during compression and expansion is assumed to be triangular in shape, find the moment of inertia of the flywheel. 18

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